

Partial correlation between paddy production and rainfall

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```
# Loading the data set:-
```

```
x = read.delim("C:/Users/rahul/OneDrive/Documents/proj4.txt")
```

```
# Knowing some primary things about the data set:-
```

```
dim(x)
```

```
## [1] 6 5
```

```
names(x)
```

```
## [1] "Rice"      "Jowar"      "Jute"      "Rainfall" "Temp"
```

```
# Computing Partial correlation:-
```

```
f3 = lm(Rice~Temp , data = x) # Regress Rice on Temp
```

```
f4 = lm(Rainfall~Temp, data = x) # Regress Rainfall on Temp
```

```
f13 = f3$residuals # Taking residuals(true value - estimated value) of f3
```

```
f14 = f4$residuals # Taking residuals(true value - estimated value) of f4
```

```
r = cor(f13,f14) # Partial correlation between Rice production and rainfall
```

```
r # Print required partial correlation
```

```
## [1] 0.1944514
```